LIMS Print Pipeline

Illustrated Operations Manual — every step, with real screenshots of the hub dashboard, the install wizard, and the print-time dialog

What this system does

Staff on 60 internet-isolated PCs print from any application. Each job becomes a PDF, is tagged with a **registration number** and **test** picked from live dropdowns, is filed into a shared folder on a central desktop, and (when a cloud LIMS is configured) is forwarded there automatically. A job without a valid registration is safely **held** until an operator assigns one. It runs entirely on the LAN — only the one central desktop needs internet.

The three machines

Tier	Machine	Internet	What runs here
Cloud LIMS	Supabase + a browser	yes	Testers create registrations; documents appear live
Central hub	one desktop e.g. 192.168.1.172	yes	hub\run.bat; hosts the limsDocs share
Clients (x60)	each lab PC	no (LAN)	install.bat wizard; prints to the hub

Everything you will ever run

Run this	Where	Purpose
`hub\run.bat`	hub	Start the hub (keep the window open)
`schema.sql (Supabase SQL editor)`	cloud	Create the LIMS database (once)
`npm install / npm run dev`	cloud	Start the tester web app
`install.bat`	each client	GUI wizard: create + enroll a printer
`add-printer.bat`	each client	GUI wizard: add another printer
`status.bat`	client	Show printers, devices, recent log
`fix-queue.bat`	client	Repair a stuck queue + diagnostics
`change-password.bat`	client	Change ONE printer's PDF password (super-admin)
`view-password.bat`	client	Reveal a printer's PDF password (super-admin)
`set-super-admin.bat`	client	Rotate the super-admin password
`set-password.bat`	client	Global PDF password (all printers)
`decode-pdf.bat`	anywhere	Decrypt an encrypted PDF
`uninstall.bat`	client	Remove printers, port, files

PART 1 Cloud LIMS — one-time setup

Do this once on any machine with a browser. Skip it to run the hub in **local mode** (no cloud) and add Supabase later — queued jobs forward automatically once configured.

Step	Action / value
1	supabase.com → create a project. Copy the Project URL, anon key, and service_role key (Settings → API).
2	SQL Editor → paste lims/supabase/schema.sql → Run (idempotent, safe to re-run).
3	Authentication → enable the Email provider; add a login (email + password) for each tester.
4	(optional) more device types: insert into device_types (id) values ('xrf');

Start the tester web app

```
cd lims\web
copy .env.example .env :: then set these two lines from Settings -> API:
VITE_SUPABASE_URL=https://YOUR-PROJECT-REF.supabase.co
VITE_SUPABASE_ANON_KEY=YOUR-ANON-KEY
npm install
npm run dev :: or npm run build to host dist\
```

Never put the service_role key in the web app

The web app uses the ANON key only (plus row-level security). The service_role key belongs solely on the central desktop, entered on the hub's Settings tab (Part 2).

What testers enter in the web app

Screen	Field	Value
Sign in	Email / Password	the tester login from step 3
Registrations	Registration number	e.g. R-2026-0001
Registrations	Device type	gcms / lcms / icpms ...
Registrations	Product	free text (optional)
Registrations	Tests	type a name, press Enter per chip (e.g. assay)
Documents	Download	filter by reg/device; opens the PDF via a signed URL

PART 2 Central hub — one-time setup

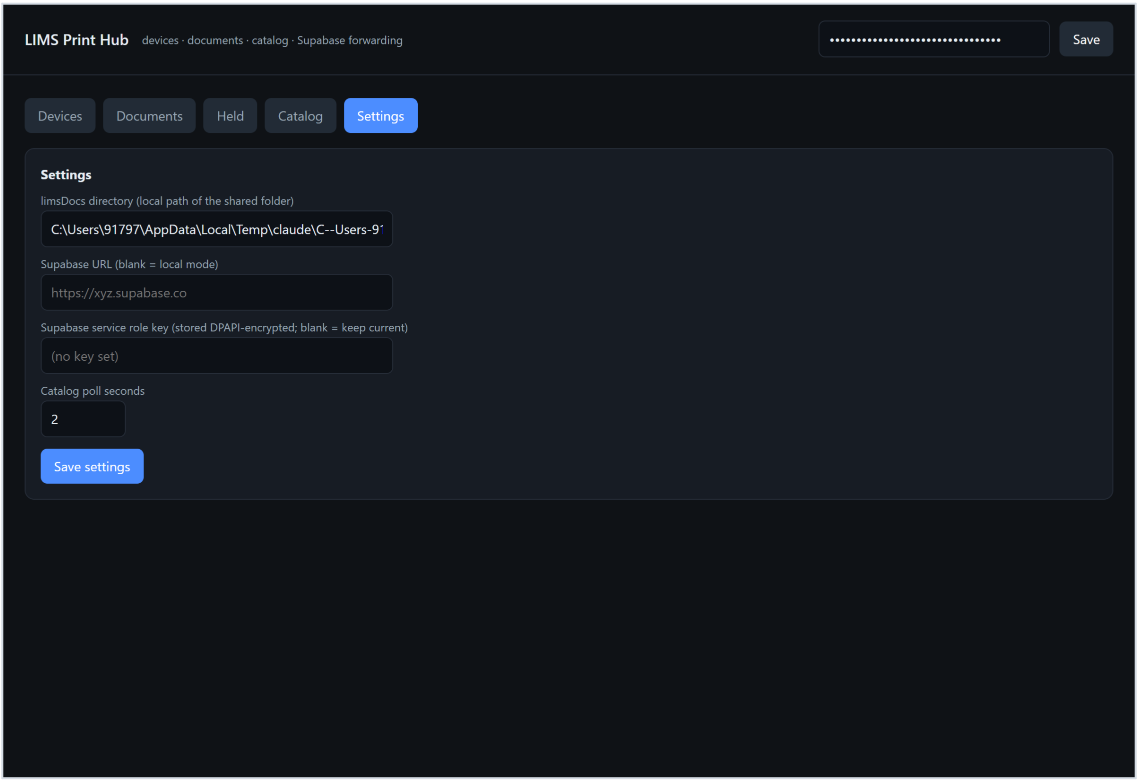
On the central desktop (the one with internet + the limsDocs share). Needs **uv** (auto-installed on first run).

Prepare the share, then start the hub

```
mkdir C:\limsDocs :: share it on the LAN, grant lab users READ
cd hub
run.bat :: double-click, or run in a Command Prompt - LEAVE IT OPEN
```

The window prints two secrets (they persist across restarts): the **ADMIN TOKEN** (unlocks the dashboard) and the **ENROLL KEY** (used by each client's wizard). Open the dashboard at `http://localhost:8000/` and paste the admin token.

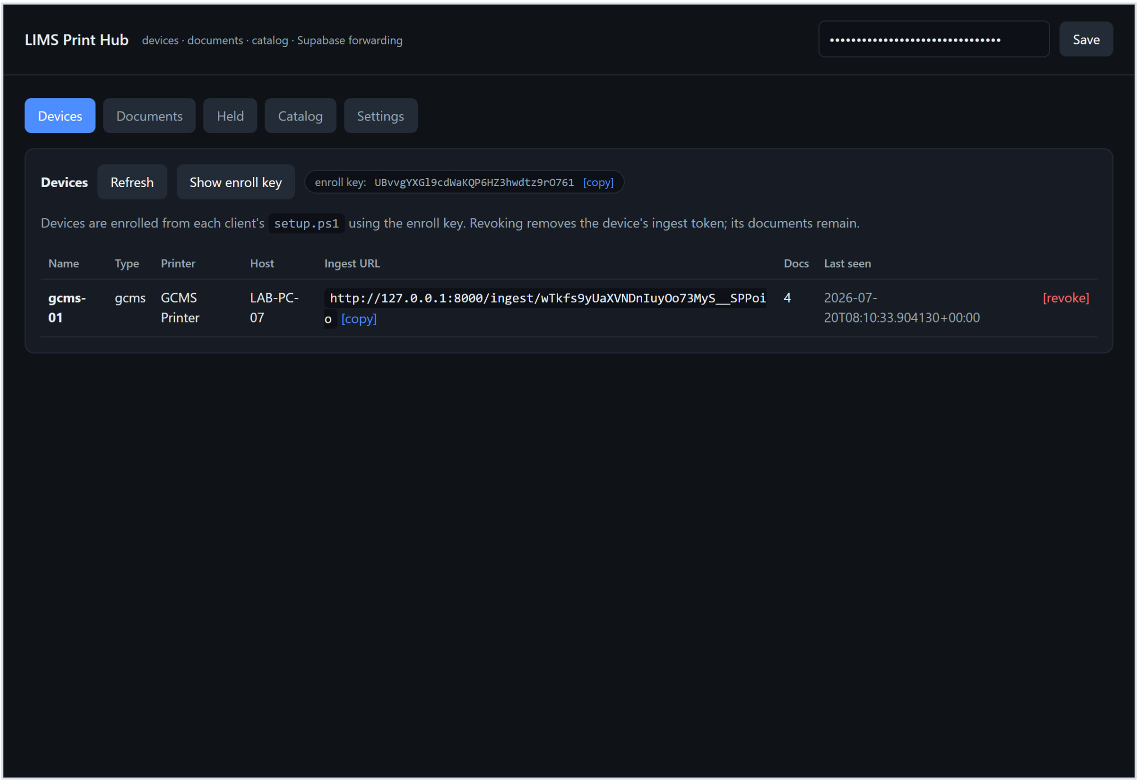
Settings tab — point the hub at the share (and, optionally, Supabase)



The Settings tab: limsDocs directory, Supabase URL + service-role key (stored DPAPI-encrypted, shown masked), and the catalog poll interval.

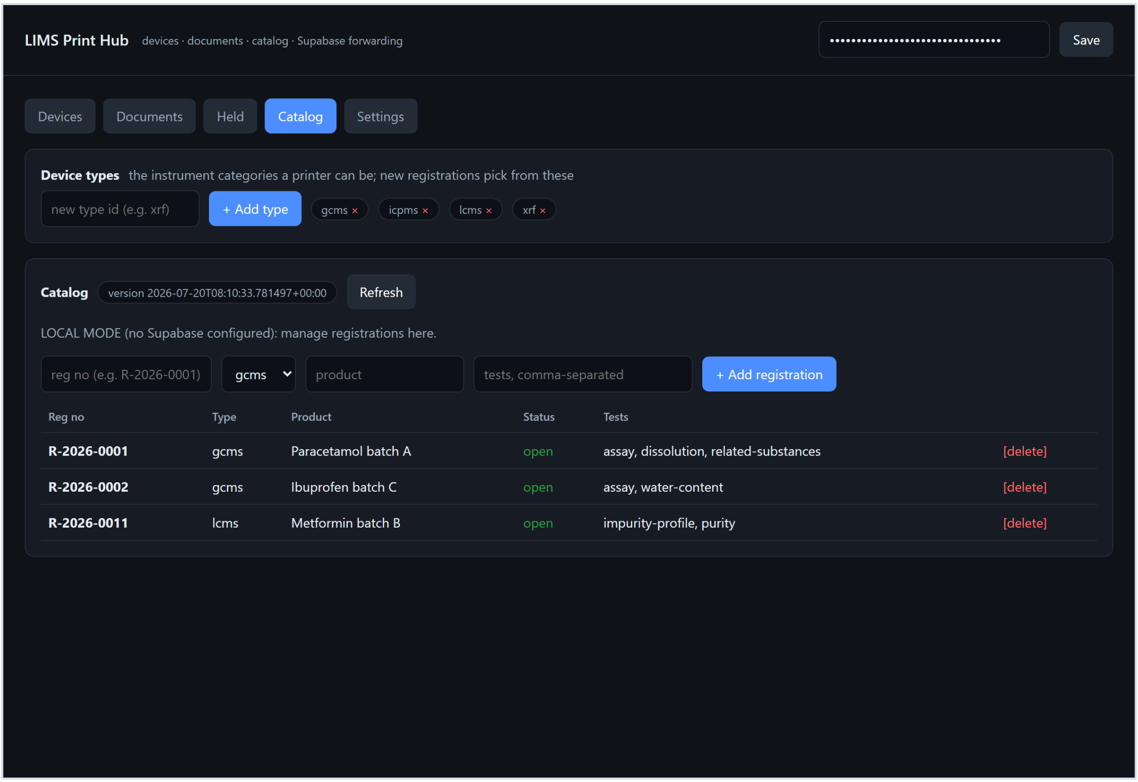
Field	Value
limsDocs directory	C:\limsDocs (the LOCAL path of the shared folder)
Supabase URL	your Project URL (blank = local mode)
Supabase service role key	the service_role key (stored DPAPI-encrypted on this machine only)
Catalog poll seconds	2 (default)

Devices tab — see enrolled printers & reveal the enroll key



Devices tab. "Show enroll key" reveals the key to give to each client's wizard. Each printer has its own ingest URL; "revoke" kills just that device's token.

Catalog tab — device types & registrations (local mode)



Catalog tab. Top: the Device types manager - type a new id (e.g. xrf) and Add it, or remove one with the × (a type still used by a registration/device can't be deleted). Below: add registrations (reg no, device type, product, comma-separated tests). With Supabase configured this mirrors the cloud and is overwritten by the next sync.

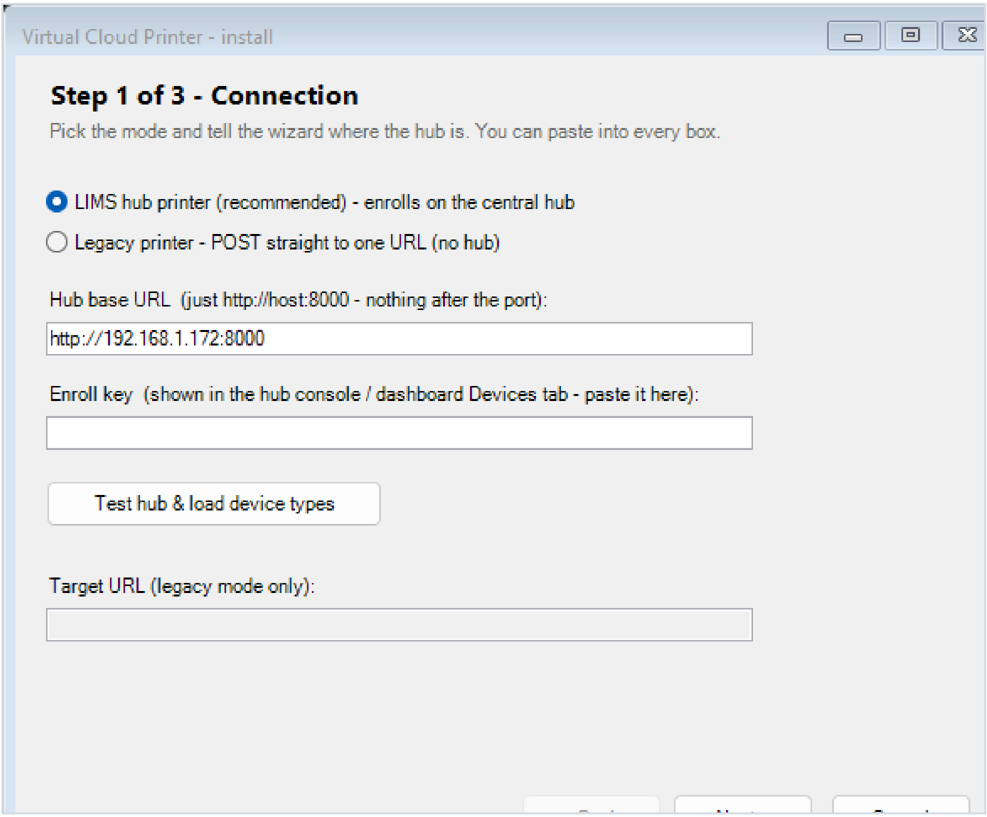
Manage	How
Add a device type	Catalog tab → type an id under "Device types" → + Add type. New registrations then offer it.

Manage	How
Remove a device type	Click the × on its chip (blocked with 409 if a registration/device still uses it).
Add a registration	reg no + device type + product + comma-separated tests → + Add registration.

Client PC — install a printer (the wizard)

On each lab PC, double-click **install.bat** and approve the UAC prompt. A click-through wizard opens — **you can paste into every box**. First run downloads Ghostscript, mfilemon and Python once.

Step 1 — Connection

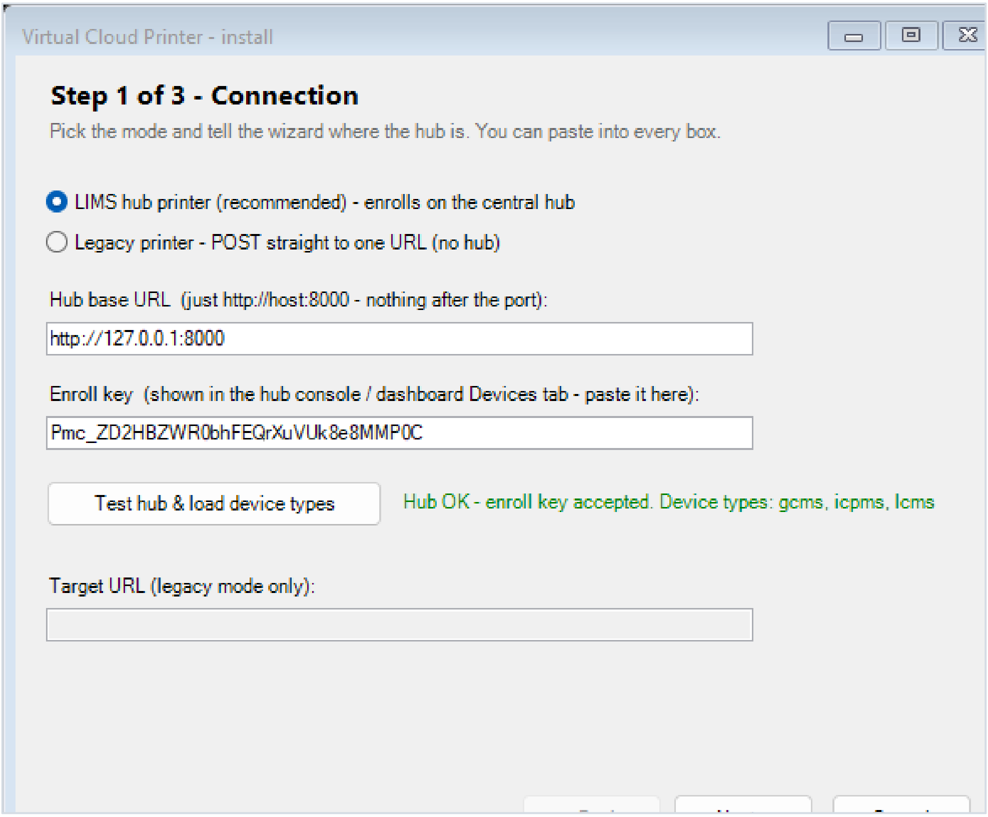


Step 1: choose "LIMS hub printer", enter the hub base URL and paste the enroll key.

The #1 rule: Hub base URL is http://HOST:8000 — nothing after the port

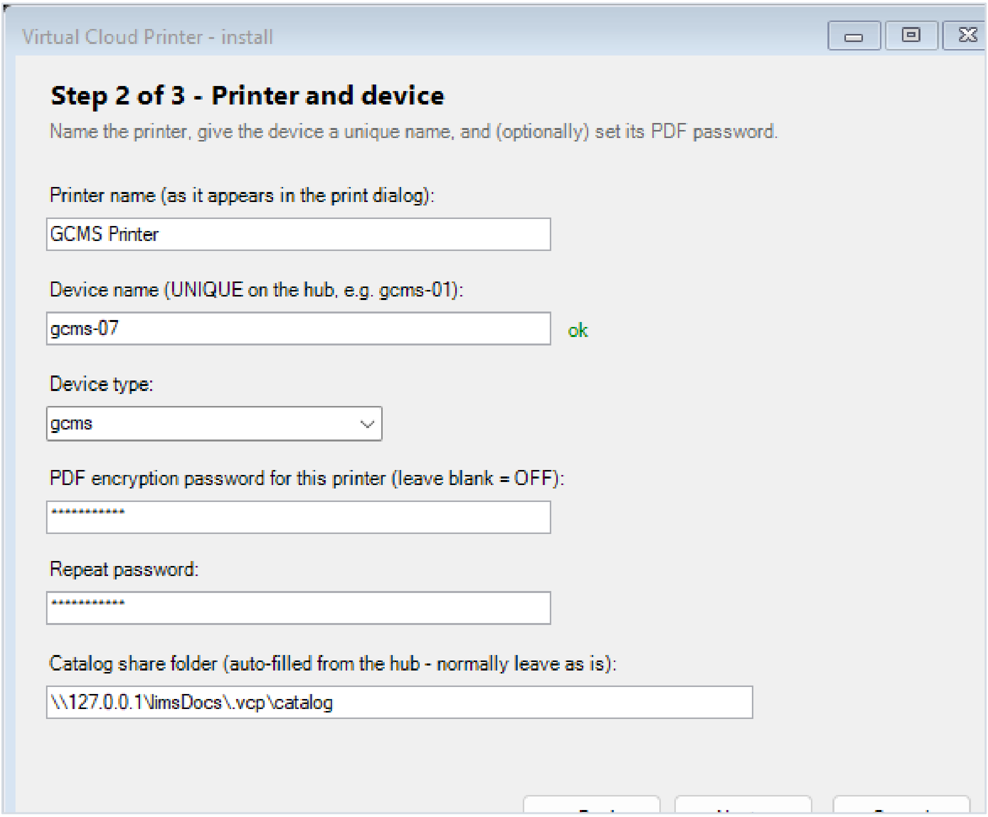
Never paste an /ingest/... link here (that is a per-printer address the hub issues automatically). The wizard rejects /ingest/ URLs.

Click **Test hub & load device types** — a green confirmation means the key is accepted and the device-type dropdown is filled from the hub:



Green "Hub OK - enroll key accepted" with the device types loaded from the hub.

Step 2 — Printer and device

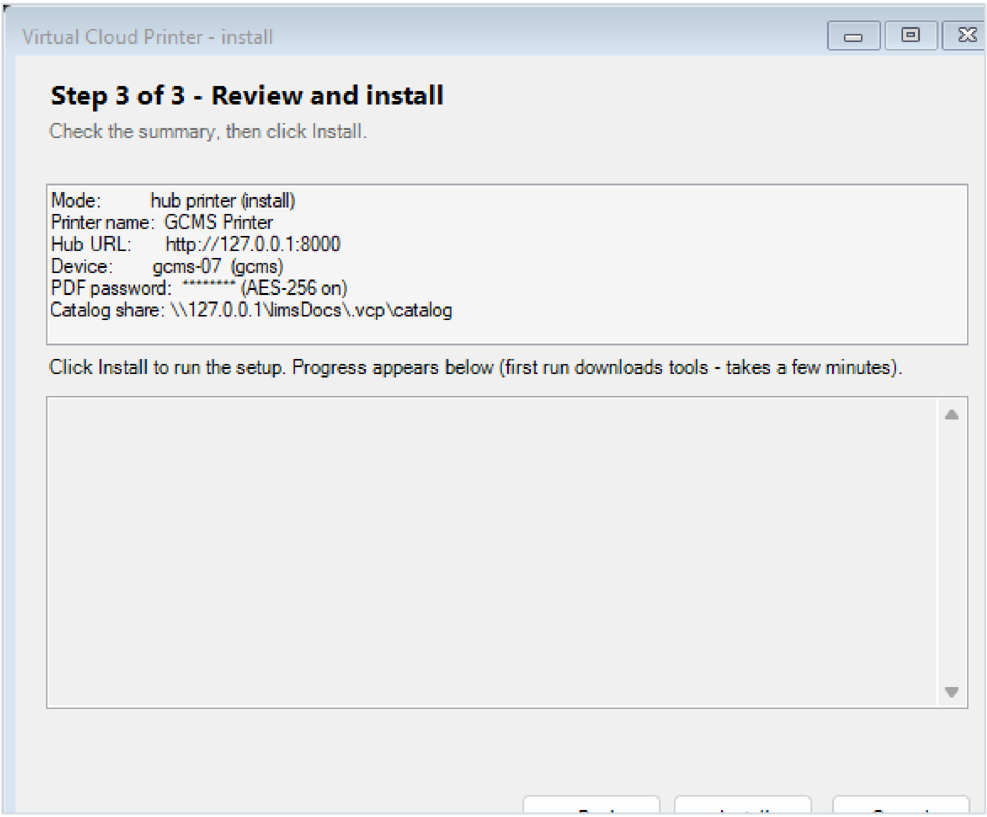


Step 2: printer name, a UNIQUE device name (live "ok" check), device type, an optional PDF password (entered twice), and the auto-filled catalog share folder.

Field	What to type	Notes
Printer name	e.g. GCMS Printer	how it appears in the Windows print dialog

Field	What to type	Notes
Device name	e.g. gcms-01	UNIQUE across all clients (409 if taken)
Device type	gcms / lcms / icpms	dropdown, loaded from the hub
PDF password	a password, or blank	blank = no encryption; stored DPAPI + via a locked temp file
Catalog share folder	leave the auto value	\\\limsDocs\.vcp\catalog

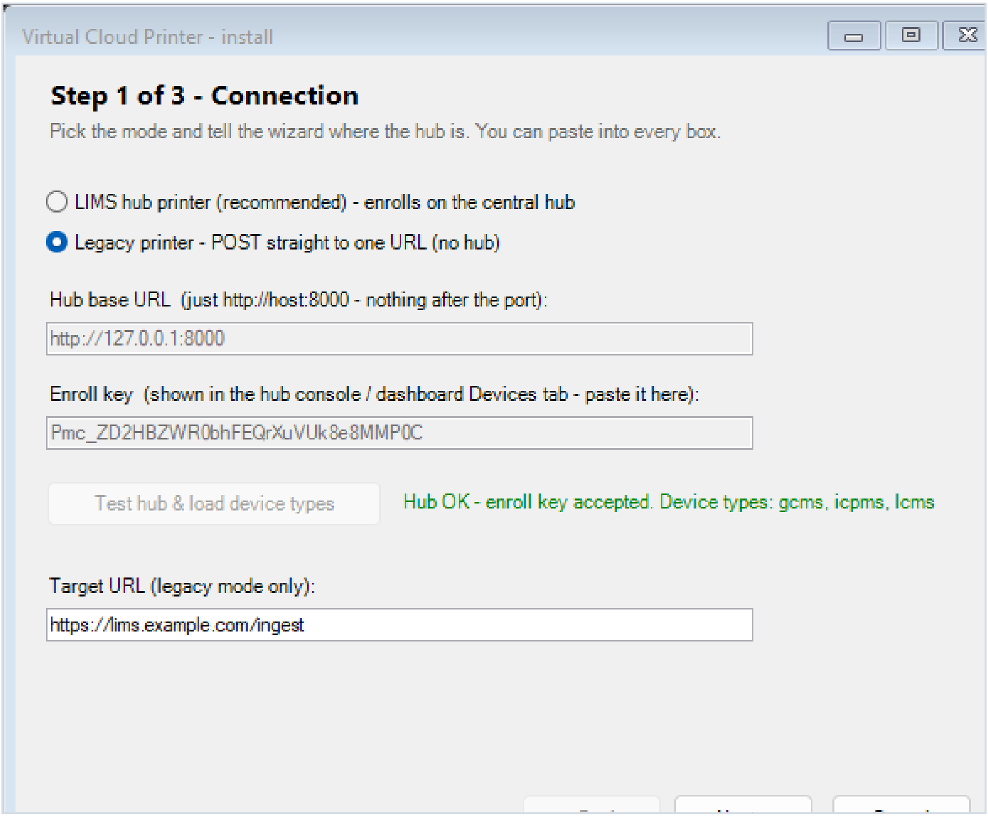
Step 3 — Review & Install



Step 3: review the summary, then click Install. Progress streams in the window; it ends with a green DONE.

Legacy (direct-URL) printers

To POST straight to one URL with no hub, pick **Legacy printer** on Step 1 and enter a Target URL:



Legacy mode: only a Target URL is needed; no device/hub fields.

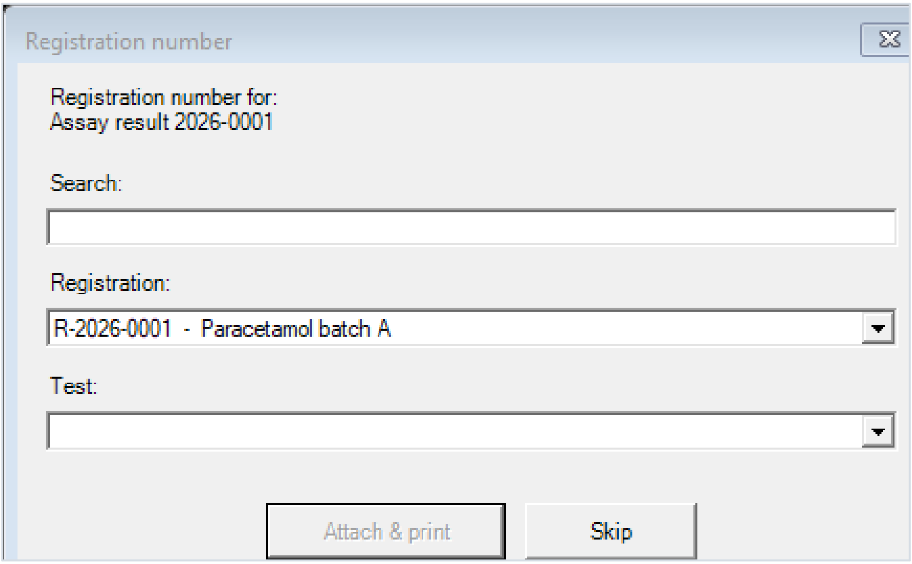
Scripted install (mass rollout, no wizard)

```
powershell -ExecutionPolicy Bypass -File setup.ps1 -Action install ^
-PrinterName "GCMS Printer" -HubUrl http://192.168.1.172:8000 ^
-DeviceName gcms-01 -DeviceType gcms -EnrollKey ^
-Password -CatalogShareDir \\192.168.1.172\limsDocs\.vcp\catalog
```

PART 4

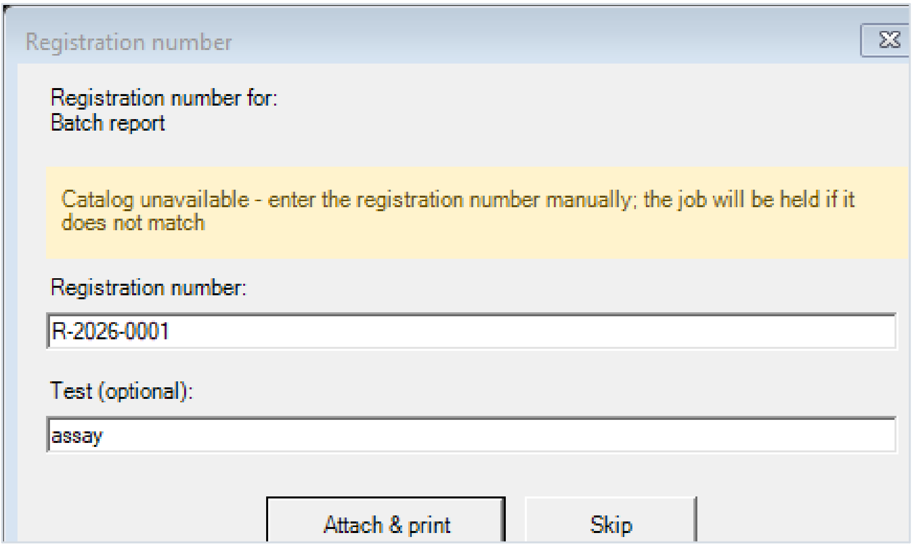
Everyday use — print, pick, file

Print from any application and choose the printer. A dialog appears on your screen. For a hub printer it shows two dropdowns fed live from the catalog:



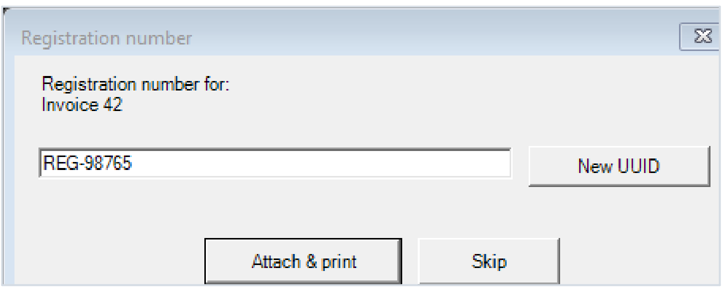
The print-time dialog (hub printer): search, pick the Registration, then its Test, then Attach & print. Skip sends the job anyway - the hub holds it until assigned.

If the catalog can't be reached (hub down and share unreadable), the dialog falls back to free text with a warning, so work never stops — the hub validates and holds if the number doesn't match:



Fallback: catalog-unavailable warning + free-text registration and test.

A legacy (direct-URL) printer shows the original free-text box with a New-UUID button:



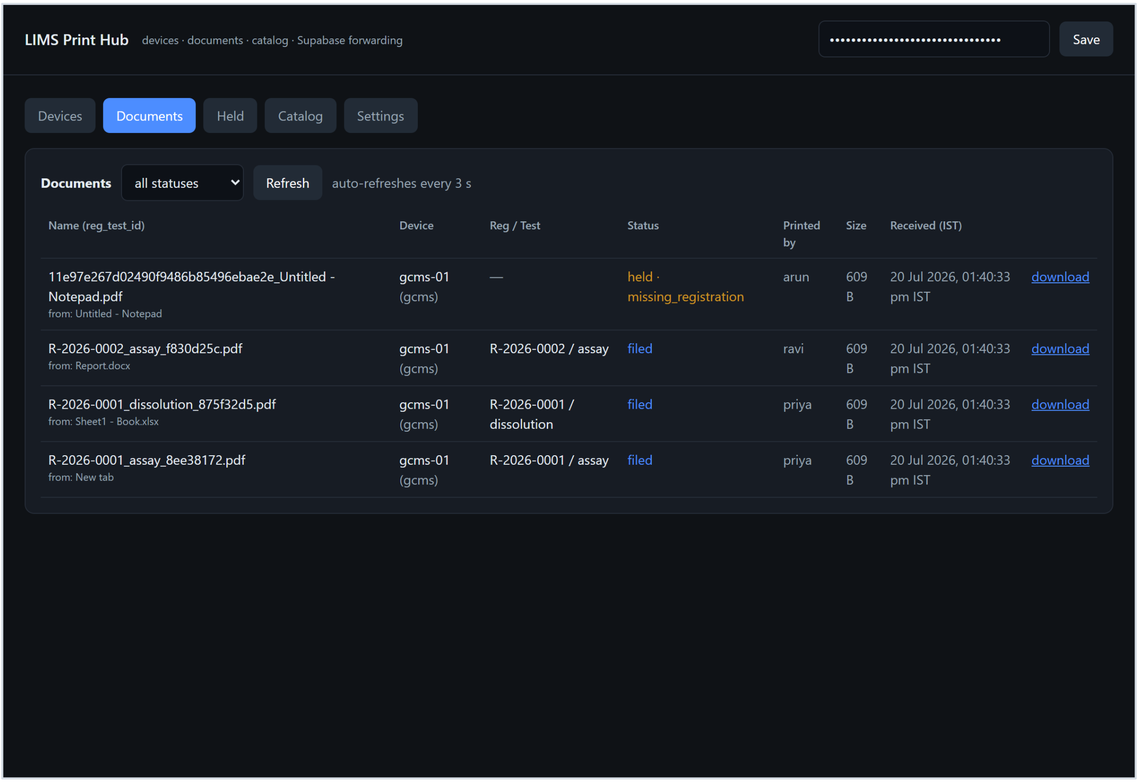
Legacy printer dialog: free-text registration / UUID.

Where the file lands & how it is named

The file is named **<registration>_<test>_<id>.pdf** — a lab-friendly identifier, not the app's print title ("New tab", "Untitled - Notepad"). The original title is kept as metadata.

```
\\192.168.1.172\limsDocs\\\\\__.pdf
:: e.g. ...\gcms-01\gcms\R-2026-0001\assay\R-2026-0001_assay_8ee38172.pdf
:: and, with Supabase configured, in the LIMS Documents page within seconds
```

The hub's Documents tab — every job, live

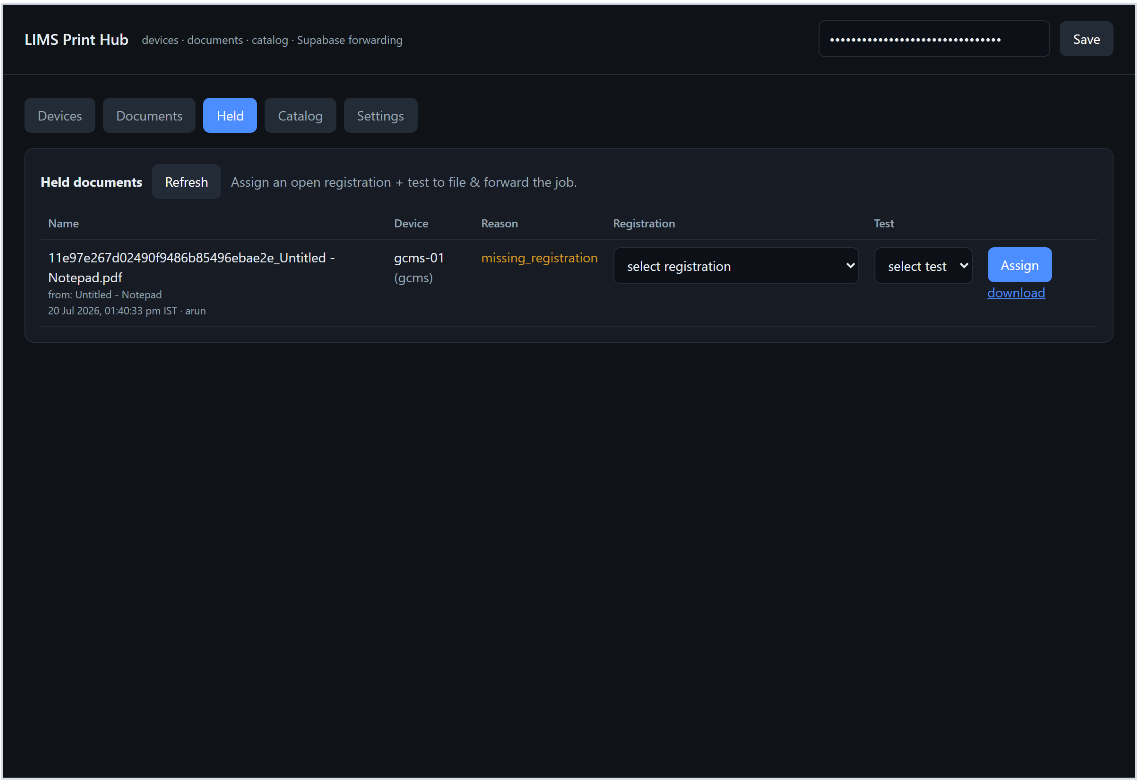


Documents tab: each row is named reg_test_id (with the original title shown as "from: ..."), plus the device (which persists even if later revoked), registration/test, status, who printed it (Printed by), size, and the time in IST. Auto-refreshes every 3 s (paused while you interact); download any PDF.

PART 5 Held documents — registration is required

A job is **held** (never lost, never forwarded) when the user clicked Skip, the dialog timed out, or the registration/test didn't validate. The reason is recorded:

Held reason	Meaning
missing_registration	no registration number was supplied
unknown_registration	the number is not an open registration
device_type_mismatch	the registration belongs to a different device type
missing_test	a registration was given but no test
unknown_test	the test is not one of that registration's tests



The Held tab: each held job shows its reason plus a Registration and Test dropdown. Pick the correct pair and click Assign - the file moves into the tree and forwards to the cloud. The 3 s auto-refresh pauses while you are choosing, so it never resets your selection.

Assigning files the job into the limsDocs tree (renamed to reg_test_id) and, if Supabase is configured, forwards it. The forward is idempotent, so a retry after a network blip never creates a duplicate cloud row.

PART 6

Passwords & security

Factory super-admin password: Citta@26252423 - rotate it on day one

Run set-super-admin.bat on each client to set your own. It is stored only as a PBKDF2-SHA256 hash and gates change-password / view-password / set-super-admin.

Password commands (run on the client, elevated)

Command	Prompts (in order)	Effect
`change-password.bat`	printer; super-admin pw; new pw (twice)	set/replace ONE printer's PDF password (DPAPI at rest)
`view-password.bat`	printer; super-admin pw	show that printer's password on screen only
`set-super-admin.bat`	current; new (twice)	rotate the super-admin password
`set-password.bat`	passphrase (blank = off)	global password for printers without their own (DPAPI)
`decode-pdf.bat`	pick file(s); password	decrypt an encrypted PDF

Built-in protections (no action needed)

Protection	How
Per-device tokens	each printer's ingest token is unique; revoke in the Devices tab
Passwords at rest	DPAPI (machine scope) on clients; PBKDF2 hash for the super-admin
Cloud key on-site	service_role key stays DPAPI-encrypted on the hub only; web app uses the anon key
No path escape	every folder segment from a client is sanitized (verified)
Fail-closed	encryption on + no password/qpdf -> job held in failed\, never sent in the clear
Idempotent forward	documents insert upserts on storage_path -> no duplicate cloud rows on retry
Secret off the cmdline	the wizard passes the PDF password via an ACL-locked file, not the process args

PART 7 Test it on ONE PC first (local mode, ~15 min)

Proves the whole pipeline on a single Windows PC with no LAN and no Supabase.

1. **Start the hub** `cd hub` then `run.bat`; note the ADMIN TOKEN + ENROLL KEY; dashboard at http://localhost:8000/.
2. **Set the folder** Dashboard → paste admin token → Settings → limsDocs = C:\limsDocs (create it) → Save.
3. **Add a registration** Catalog tab → add R-TEST-1, type gcms, tests assay, dissolution.
4. **Install a printer** install.bat → wizard: hub http://127.0.0.1:8000 + enroll key → Test hub → Next → printer limsPrinter, device gcms-01, type gcms, blank password → Next → Install.
5. **Print** Notepad → Print → limsPrinter → the dialog shows R-TEST-1 → pick it → assay → Attach & print.
6. **Verify** C:\limsDocs\gcms-01\gcms\R-TEST-1\assay\ has the PDF; the Documents tab shows it as filed.
7. **Safety net** Print again and click Skip → the Held tab shows it; Assign a registration + test to file it.

If the dialog doesn't appear or nothing files

Run fix-queue.bat (elevated) - it repairs permissions, restarts the spooler, and writes vcp-diagnostics.txt. Read C:\ProgramData\VirtualCloudPrinter\log.txt for the per-job trace. Unposted PDFs are always kept in failed\.

PART 8

Maintenance & troubleshooting

Symptom	Cause & fix
Dropdown is empty	No open registration of that device type - add one (LIMS / Catalog tab). Check the printer's type with status.bat.
"Catalog unavailable"	Client can't reach the hub AND the share fallback is unreadable - check the hub is up, port 8000, and the .vcp\catalog files exist.
Enroll fails 409	Device name already used - pick a unique one.
Enroll fails 401	Wrong enroll key - Devices tab -> Show enroll key.
Wrong app on :8000	Title says "Receiver Simulator" = the old dev tool (now on 8001). Run hub\run.bat instead.
Document stuck 'held'	Expected without a valid registration - assign it in the Held tab.
'forward_failed'	Hub can't reach Supabase - check internet + keys in Settings; it retries automatically.
Nothing prints	Run fix-queue.bat; read log.txt; check the client's failed\ folder.

Routine checks

Run / open	Shows
`status.bat (client)`	monitor, printers, device names/types, last log lines
`Hub -> Documents`	live job statuses (filed / held / forwarded / forward_failed)
`Hub -> Devices`	enrolled clients + last-seen; revoke a token
`C:\ProgramData\VirtualCloudPrinter\log.txt`	per-job client trace

Companion documents

LIMS-Pipeline-Reference.pdf (architecture + flow diagrams + security) - LIMS-Pipeline-Operations-Manual.pdf (text command reference) - LAN-SETUP.md / ARCHITECTURE.md (in the repo). Every screenshot here is a real capture of the running hub dashboard and the actual install/print dialogs.